P-D-425c
August 18, 1966

SUPERSEDING
Int. Fed. Spec. P-D-00425b (NAVY-SHIPS)
May 7, 1965 and
Fed. Spec. P-D-425a
May 4, 1954

FEDERAL SPECIFICATION

DISHWASHING COMPOUND, MACHINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers free-flowing, solid form dishwashing compounds suitable for use in spray-type mechanical dishwashing machines.

1.2 Classification. Dishwashing compound shall be of the following types, as specified (see 6.2):

Type I—For hard water.
Type II—For soft water.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).
QQ-A-250/2—Aluminum Alloy 3003, Plate and Sheet.
RR-S-366—Sieve, Test.
UU-S-48—Sacks, Shipping, Paper.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-655—Boxes, Fiberboard, Six or Eight Sides.
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents): Sampling and testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material. Dishwashing compound shall be formulated from alkaline salts, sequestering agents in proper proportion to provide a compound which will meet all the requirements of this specification. No agents shall be used which contribute to excessive foaming under conditions of use.

3.2 Odor. Dishwashing compound shall be free from objectionable odor in dry form.

3.3 Foaming. When tested as specified in 4.4.4, the compound shall not foam to the extent as to interfere with soil removal and rinsing of utensils and dishes during dishwashing.

3.4 Moisture content. The moisture content of the compounds (types I and II), shall not exceed 8 percent.

3.5 Alkaline salts. The alkaline salts content of the compounds (types I and II), shall be at least 30 percent (expressed as sodium oxide, Na_2O).

3.6 Total phosphates (as P_2O_5). Dishwashing compound shall contain complex alkali phosphate-equivalent to 27 percent minimum P_2O_5 for type I and 20 percent minimum P_2O_5 for type II.

3.7 Total silicates (as SiO_2). Type I compound shall have at least 10.0 percent silicates (as SiO_2) and type II compound shall have at least 15.0 percent silicates (as SiO_2).

3.8 pH and buffer capacity. When tested as specified in 4.4.9, a solution of the compound shall have an initial pH of not less than 10.5 and not higher than 12.5. The addition of the equivalent of 10.0 milliliters (ml.) of 0.1 *N* hydrochloric acid shall not reduce the pH of the solution to less than 9.0.

3.9 Water softening capacity. When tested as specified in 4.4.10, the hard water shall be softened completely as indicated by the absence of turbidity.

3.10 Turbidity. When tested as specified in 4.4.11, the weight of precipitate obtained shall not exceed 15 milligrams.

3.11 Corrosion. When tested as specified in 4.4.12, aluminum test specimens shall not be discolored or etched, nor shall a dense white film be formed on the surfaces. Slight dulling of the surfaces or formation of a faint white film on the test specimens shall not be interpreted as evidence of nonconformity with this requirement.

3.12 Matter insoluble in distilled water. Matter insoluble in distilled water shall not exceed 1 percent by weight when determined as specified in 4.4.13.

3.13 Toxicity. The ingredients used in formulating dishwashing compound shall not be contaminated with toxic amounts of poisonous compounds such as those of arsenic, lead, mercury, antimony, or any other materials having a deleterious effect. The dishwashing compound shall contain not more than 0.003 percent heavy metals expressed as lead, when tested as specified in 4.4.14.

3.14 Package stability. When tested as specified in 4.4.15, not more than 5 percent of the compound shall be retained on a No. 10 standard sieve.

3.15 Fineness. The particle size of mechanical mixture of ingredients shall be normally uniform in order to minimize segregation of the ingredients. When tested as specified in 4.4.16, not less than 95.0 percent of the material shall fall within the following range:

Passing	Retained
No. 16	No. 170

3.16 Labeling.

3.16.1 *Type I.* Unless otherwise specified in the contract or order, each container of type I compound shall be durably and legibly labeled with black ink with the following information:

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Dishwashing Compound for Hard Water (above 120 parts per million (p.p.m) hardness) P-D-425c, type 1.

Density—ounces per 8 fluid ounce level cupful (to be inserted by contractor).

INSTRUCTIONS FOR USE

For water containing hardness (as calcium carbonate) of the range above 7 to 14 grains per gallon (above 120 to 240 p.p.m.), use 5 ounces of type I compound for each 10 gallons of wash water. For water containing hardness in the range above 14 to 20 grains per gallon (above 240 to 342 p.p.m.), use 7 ounces for each 10 gallons of water.

To determine the amount needed, multiply the indicated dosage at the predetermined hardness by the number of gallons of wash tank capacity and divide by 10.

If water hardness is 7 grains per gallon or less, use type II compound.

During operation of a dishwashing machine not equipped with automatic dispenser or a detergent meter, the compound should be replenished by adding about one-fourth of the original amount every 10 minutes while tableware is being washed.

Machine should be drained after 1-1/2 to 2 hours of operation in order to prevent foaming and insure maximum cleaning of tableware.

3.16.2 *Type II.* Unless otherwise specified in the contract or order, each container of type II compound shall be durably labeled with black ink with the following information:

Dishwashing Compound for Soft Water (from 0 to 120 p.p.m. hardness) P-D-425c, type II.

Density—ounces per 8 fluid ounce level cupful (to be inserted by contractor).

INSTRUCTIONS FOR USE

Three and one-half ounces of type II compound should be added to each 10 gallons of wash water having a hardness (as calcium carbonate) at 7 grains per gallon (120 p.p.m.) or less. To determine the amount needed, multiply 3-1/2 ounces by the number of gallons of wash tank capacity and divide by 10. If water hardness is above 7 grains per gallon use type 1 compound. During operation of a dishwashing machine not equipped with an automatic dispenser or a detergent meter, the compound should be replenished by adding about one-fourth of the original amount every 10 minutes while tableware is being washed.

Machine should be drained after 1-1/2 to 2 hours of operation in order to prevent foaming and insure maximum cleaning of tableware.

3.17 Workmanship. The material shall be thoroughly mixed and present no evidence of segregation of individual constituents or of lumping or caking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, a lot for inspection and tests shall consist of all material of the same type submitted for inspection at the same time.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQLs) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of containers, as specified, of the applicable quantity and type material contained therein.

4.3.1.1 *Examination of the filled container for defects in design and construction.* The sample unit for this examination shall be one filled container.

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Examine	Defect
Container	Not type specified.
	Any split, break, dent, hole, tear, puncture, or other defect in sidewall which may effect serviceability of container.
	Any evidence of seepage.
	Top and bottom ends of sack not dipped.
	Thickness of filled sack more than 30 percent of its width when laid flat.
Markings	Not in accordance with contract requirements.
	Illegible, incorrect, or incomplete.
	Label, lithograph, printing, or stenciling pertaining to the applicable information and instructions for end item use, missing, illegible, incomplete, or smears.
	Paper label (when used) not securely affixed; not coated overall with a transparent lacquer or enamel.
	Not in one print black ink.

4.3.1.2 *Examination of the finished product for defects in workmanship and interior pack.* The sample unit for this examination shall be one filled container.

Examine	Defect
Workmanship	Not uniform.
	Not free flowing; not powder, granule, nor flake.
	Any evidence of segregation of constituents; lumping or caking.
	Any objectionable odor.
Interior pack (as applicable)	Folding cartons (when specified), not lined with liner material.

4.3.1.3 *Examination of the filled containers for determining compliance with the requirements for net contents.* The sample unit for this examination shall be one filled container. The average net content per container shall be not less than the specified or indicated quantity.

4.3.1.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and markings comply with section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. This examination is not applicable when the primary container is the shipping container. The lot size shall be expressed in units of shipping containers in the end item inspection lot.

Examine	Defect
Packaging (as applicable)	Not in accordance with contract requirements.
	Unit package not packaged as specified.
	Packaging material not as specified.
Packing (as applicable)	Not in accordance with contract requirements.
	Arrangement or number of unit cartons per container not in accordance with contract requirements.
	Inadequate application of components such as: incomplete closure of case liners; container flaps, loose or inadequate strappings; bulged or distorted containers.
	Container not as specified; closures not accomplished by specified or required methods or materials.
	Liner material (as applicable), omitted, damaged, or otherwise defective, affecting serviceability.
Weight	Gross or net weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable), omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.3.1.5 *Inspection levels and AQLs for examinations.* The inspection levels, for determining the sample size, and the AQLs, expressed in defects per 100 units, as applicable, shall be as follows:

Examination par.	Inspection levels	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-2	4.0
4.3.1.3	S-2	—
4.3.1.4	S-2	4.0

4.3.2 *Testing of the end item.* The end item shall be tested for characteristics as in-

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dicated in table I for each lot presented for examination for each type of material. The sample unit, for purposes of testing, shall be one pound composite sample selected in accordance with 4.3.2.1. Unless otherwise specified, two determinations shall be performed on the composite sample. The average of two determinations shall meet the requirements when results are reported numerically. Both determinations shall meet the requirements when the test is applicable to the individual unit. The lot size shall be expressed in units of containers of the specified quantity and type of dishwashing compound.

4.3.2.1 *Composite sample.* A sample of not less than one pound each shall be taken from not less than four separate shipping containers, randomly selected throughout the lot presented for test. The total samples shall, in all cases, consist of not less than three sample portions taken at random from each separate shipping container, and shall be placed in a container of a suitable size. After thorough mixing, a one pound sample shall be drawn for testing purposes. The composite sample shall be placed in an airtight container and sealed, and marked with the identifying lot number and date.

4.4 Test procedures.

4.4.1 *Preparation of laboratory sample.* Samples shall be quartered prior to analysis.

4.4.2 *Preparation of stock solution of sample.* Prepare a solution of the compound by dissolving 50.0 grams in CO_2-free[1] distilled water at 55°-65°C. Cool to 20° ± 1°C., make up to exactly 1 liter with water and mix thoroughly. This solution shall be allowed to stand for not more than 1 hour to allow any insoluble matter to settle. Approximately 500 ml. of the clear supernatant solution shall be decanted into a clean dry, glass-stoppered bottle. If the solution does not settle rapidly, it may be filtered through a dry filter, the first 100 ml. of the filtrate discarded, and the next 500 ml. collected in the bottle.

4.4.3 *Preparation of stock solution of hard water.* Prepare a solution (using CO_2-free[1] distilled water) containing calcium chloride and magnesium chloride equivalent to 2.85 mg. of calcium carbonate and 1.20 mg. of magnesium carbonate per ml., respectively. This solution contains 4275 p.p.m. hardness (approximately 250 grains per gallon) as calcium carbonate. It should be prepared by dilution of more concentrated stock solutions of calcium chloride and magnesium chloride which have been accurately analyzed. The deviation from the specified concentrations should not exceed 1 part per 100.

4.4.4 *Foaming.* Test shall be performed in a commercial tank dishwashing machine. During operation of the machine, the dishwashing compound concentration shall be maintained in accordance with instructions for use in 3.16.

4.4.5 *Moisture content.* Determine the moisture content in accordance with method 201.1 of Fed. Test Method Std. No. 536.

4.4.6 *Alkaline salts.* Determine the alkaline salts content in accordance with method 304.3 of Fed. Test Method Std. No. 536.

4.4.7 *Total phosphates (as P_2O_5).* Determine total phosphates in accordance with method 1501.1 of Fed. Test Method Std. No. 536.

4.4.8 *Total silicates (as SiO_2).* Determine total silicates in accordance with method 1301 of Fed. Test Method Std. No. 536.

4.4.9 *pH and buffer capacity.*

4.4.9.1 *Apparatus.* Tests shall be made at 25° ± 1°C. with a pH meter having a sensitivity and readability of 0.05 pH unit. A sealed type glass electrode shall be used in conjunction with a potassium chloride-calo-

[1] CO_2-free distilled water shall be distilled water which has been freshly-boiled and cooled to 20° ± 1° C.

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TABLE I. *Instructions for testing*

Characteristic	Specification reference		Rqmts. appl. to		Number determinations per composite sample unit	Results reported as	
	Requirement	Test method	Indiv. unit	Lot average		Pass or fail[1]	Numerically to nearest[2]
Foaming	3.3	4.4.4	X	—	1	X	—
Moisture content	3.4	4.4.5	X	—	2		0.1 percent
Alkaline salts	3.5	4.4.6	—	X	Avg. of 2		0.1 percent
Phosphates (as P_2O_5)	3.6	4.4.7	—	X	Avg. of 2		0.1 percent
Silicates (as SiO_2)	3.7	4.4.8	—	X	Avg. of 2		0.1 percent
pH and buffer capacity: Initial value	3.8	4.4.9	—	X	Avg. of 2	—	0.1 pH
pH value of acidified mixture	3.8	4.4.9	—	X	Avg. of 2	—	0.1 pH
Water softening capacity	3.9	4.4.10	X	—	2	X	—
Turbidity	3.10	4.4.11	—	X	Avg. of 2	—	mg.
Corrosion	3.11	4.4.12	X	—	2	X	—
Matter insoluble in distilled water	3.12	4.4.13	—	X	Avg. of 2	—	0.01 percent
Toxicity (heavy metals)	3.13	4.4.14	X	—	2	X	—
Package stability	3.14	4.4.15	—	X	Avg. of 2	—	0.1 percent
Fineness	3.15	4.4.16	X	—	2	—	0.1 percent

[1] If failure is indicated, report either description of fa ure or numerical point of failure, as applicable.
[2] Test reports shall include all values on which results are based.

mel reference electrode. The pH meter shall be calibrated before use by checking against a standard buffer solution of pH 10.

4.4.9.2 *Procedure.*

4.4.9.2.1 *Type I compound.* Into a 250 ml. volumetric flask, transfer 13.75 ml. of the stock solution of the sample (see 4.4.2), dilute to mark with distilled water, and mix thoroughly. Determine the pH of the solution of 25° ± 1°C. by means of a pH meter. Pipette 50 ml. of this solution into a beaker, add 10.0 ml. of 0.1*N* hydrochloric acid to the beaker, stir the solution well, and determine the pH of this acidified mixture. If the acid is stronger than 0.1*N*, less than 10.0 ml. of it therefore being required, sufficient distilled water shall be added to the beaker containing the dishwashing compound solution so that the total volume of acid and water equals 10.0 ml.

4.4.9.2.2 *Type II compound.* Type II compound shall be tested as specified in 4.4.9.2.1 except that 13.13 ml. of the stock solution shall be used.

4.4.10 *Water softening capacity.*

4.4.10.1 *Preparation of solutions for test.*

4.4.10.1.1 *Type I compound.* Transfer exactly 20 ml. of stock hard water (see 4.4.3) to a 250-ml. volumetric flask and exactly 14 ml. to another 250-ml. flask. These flasks, when subsequently filled to mark, will contain 20 and 14 grains per gallon hardness, respectively. Add approximately 100 ml. of distilled water to each flask and then 26.25 ml. of the stock solution of compound (see 4.4.2) to the first flask and 13.75 ml. to the second flask. The stock solution of the compound shall be prepared less than 60 minutes prior to the test, and shall be measured by means of a burette. Fill the flasks to mark with distilled water at 20° ± 1°C. and mix thoroughly.

4.4.10.1.2 *Type II compound.* Type II compound shall be prepared as specified in 4.4.10.1.1 except that 7 ml. of stock hard water (see 4.4.3), and 13.13 ml. of stock solution of compound (see 4.4.2), shall be used.

4.4.10.2 *Procedure.* Pipette 10 ml. of each solution prepared as specified in 4.4.10.1.1 or 10 ml. of the solution prepared as specified in 4.4.10.1.2 into a clean, dry 20 milli-

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meter by 150 mm. test tube. Into another clean, dry test tube, pipette 10 ml. of a solution prepared exactly as specified in 4.4.10.1.1 or in 4.4.10.1.2 except that the hard water is omitted. Add 5 drops (0.10 ± 0.02 ml.) of standard soap solution[2] to the 7 and 14 grains per gallon solution, or 10 drops (0.20 ± 0.02 ml.) to the 20 grains per gallon solution and swirl the tubes gently for 30 seconds. Immerse the test tubes in a distilled water bath maintained at 75° to 80°C. for exactly 5 minutes. Remove the test tubes from the water bath, and without agitation, examine for turbidity within 5 seconds after removal from the bath, using the blank as a standard. The solution shall be examined by looking through the tube in a direction perpendicular to the length of the tube. The absence of turbidity indicates that the water has been softened completely.

4.4.11 *Turbidity.* Prepare a solution of dishwashing compound in hard water as specified in 4.4.10.1.1 or 4.4.10.1.2, depending on the type of compound being tested. Maintain the solution at 75°C. for 60 minutes. If no turbidity is noted, report "turbidity none". If turbidity is present, filter through a weighed Gooch crucible, wash the crucible twice with small portions of hot water, and then dry to constant weight at 105° C. Report weight of this precipitate as "milligrams of turbidity".

4.4.12 *Corrosion.* Totally immerse duplicate specimens (approximately 3 inches by 0.75 by 0.064 inch) of bright-finished, uncoated, aluminum-alloy sheet, conforming to temper H14 of QQ-A-250/2 at 82°C. (180° F.) for 5 hours in 250 ml. solutions of the compound prepared as specified in 4.4.12.1 or 4.4.12.2. Four-hundred-ml. beakers are of convenient size for the solutions. Maintain solution level with distilled water. After 5 hours', exposure, remove the specimens, rinse with distilled water, and dry. Examine the specimens for discoloration, etching, or formation of a white film on the surface.

4.4.12.1 *Type I compound.* Into a 250-ml. volumetric flask, transfer 18.75 ml. of the stock solution of sample (see 4.4.2), adjust to 20° ± 1°C., dilute to mark with distilled water, and mix thoroughly.

4.4.12.2 *Type II compound.* Into a 250-ml. volumetric flask, transfer 13.13 ml. of the stock solution of sample (see 4.4.2), adjust to 20° ± 1°C., dilute to mark with distilled water, and mix thoroughly.

4.4.13 *Matter insoluble in distilled water.* A sample of approximately 2.5 g. (weighed to the nearest centigram) shall be dissolved in 500 ml. of warm (55° to 65°C.) distilled water. The solution shall be permitted to stand for 15 minutes with occasional stirring. If appreciable turbidity is present, or undissolved matter remains, the solution shall be filtered through a tared Gooch crucible. The beaker and the residue on the filter shall be washed with five 10-ml. portions of warm (55° to 65°C.) distilled water, the residue being transferred quantitatively to the filter. The crucible and contents shall be dried to constant weight at 100° to 150°C., and the percentage of insoluble matter calculated. In the event that the sample dissolves completely without appreciable turbidity, filtration and weighing may be omitted and insoluble matter reported as "not appreciable".

4.4.14 *Toxicity.*

4.4.14.1 *Reagents.*

4.4.14.1.1 *Standard lead solution.* Transfer 0.159 ± 0.001 g. of c.p. lead nitrate to a 1000-ml. volumetric flask. Dilute to mark with distilled water and mix well. Pipette 10 ml. of this solution into a 100-ml. volumetic flask, add distilled water to the mark, and mix well. This solution contains 0.01 mg. lead per ml.

4.4.14.1.2 *Sodium hydroxide (app. 2N).* Dissolve 80.0 g. of c.p. sodium hydroxide pellets in somewhat less than 1000 ml. of

[2] APHA soap solution. See "Standard Methods for the Examination of Water and Sewage, 9th edition, American Public Health Association". (Note that ethyl alcohol is specified.)

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distilled water, transfer to a 1000-ml. volumetric flask, make up to mark, and mix well.

4.4.14.1.3 *Phenolphthalein indicator.* Dissolve 1.0 g. of phenolphthalein powder in 50 ml. of 95 percent ethyl alcohol. Add 50 ml. of distilled water and mix.

4.4.14.1.4 *Acetic acid (app. 1M).* Dilute 57 ml. of glacial acetic acid to 1 liter in a volumetric flask.

4.4.14.1.5 *Hydrogen sulfide water.* Pass hydrogen sulfide into distilled water for 15 minutes.

4.4.14.2 *Procedure.* Pipette 100 ml. of unfiltered stock solution of dishwashing compound (see 4.4.2) into a 250 ml. beaker. Slowly add 10 ml. of concentrated hydrochloric acid to the solution. Evaporate the solution to complete dryness on a steam bath. Bake the residue in a circulating oven at 105°C. for 2 hours. Add 50 ml. of distilled water to the residue and heat briefly on the steam bath to dissolve all soluble salts. Filter the solution quantitatively through a No. 41 Whatman filter paper. (All the insoluble silica need not be transferred to the filter paper.) Wash the beaker twice with 5 ml. portions of distilled water and pour through the filter. Finally, wash the filter paper with 5 ml. of distilled water. The washings should be collected in the beaker containing the filtrate. Add three drops of phenolphthalein indicator to the filtrate and titrate to the end-point with 2*N* sodium hydroxide. Record the volume of sodium hydroxide used. Add 1M acetic acid dropwise until the color of the indicator just disappears, then add 0.2 ml. in excess. Cover the beaker and set the solution aside. Prepare a control solution as follows: Add 15 ml. of standard lead solution plus a volume of the 2*N* sodium hydroxide equal to that used to titrate the sample to a 250-ml. beaker. Make up to 100 ml. with distilled water and add 10 ml. of concentrated hydrochloric acid. Evaporate the solution to dryness and proceed as for the sample solution except that oven baking time may be reduced to about 30 minutes. Refilter the sample solution if it is not clear, keeping the volume less than 90 ml. Transfer the sample solution and the control solution to separate 100-ml. Nessler tubes. Add 10 ml. of freshly prepared hydrogen sulfide water to each. Dilute to the mark with distilled water, stopper, and mix well. The color in the sample solution should not exceed that in the control. The color comparisons shall be made in the shortest possible time (5 minutes maximum) after the hydrogen sulfide water has been added.

4.4.15 *Package stability.* Place 250 g. of dishwashing compound in a 400-ml. low form beaker. Place sufficient lead shot in a 250-ml. low form beaker so that the combined weight of the beaker and shot equals 1,000 g. Place the weighted 250-ml. beaker on top of the 250 g. of dishwashing compound. Place the assembly in a noncirculatory oven and maintain at 130°F. for 48 hours. Remove beakers from oven and place beaker containing the compound in a desiccator at room temperature for 2 hours. Turn beaker upside down on a No. 10 mesh screen conforming to RR-S-366, remove beaker, shake screen for 2 minutes, and determine amount of compound retained on the screen. If the compound is caked in the beaker so that it does not pour, it may be loosened by tightly running the edge of a small spatula around the inside face of the beaker. Calculate percent of compound retained on the screen.

4.4.16 *Fineness.*

4.4.16.1 *Apparatus.* Test shall be made using a single-eccentric type mechanical shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute.[3] Sieves shall conform to RR-S-366.

4.4.16.2 *Procedure.* Place a 100-g. sample

[3] Rotap sieve shaker is satisfactory.

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on coarser screen of the nest of appropriate sieves and screen in shaking apparatus. Continue screening for 5 minute intervals until the weight retained on finer screen is reduced by not more than 0.1 g. during successive intervals. Weigh material on finer screen and calculate percent of material falling within the appropriate range.

NOTE. In order to evaluate results properly, excessive humidities should be avoided. In the event of failure, retests shall be conducted at a relative humidity not exceeding 50 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* The dishwashing compound shall be packaged in 2-pound, 50-pound, 100-pound, or 300-pound quantities, as specified (see 6.2), in containers as specified in 5.1.1.1 through 5.1.1.4, as applicable.

5.1.1.1 *Two pounds.* Container shall be a folding paperboard box, conforming to style I of PPP-B-566. Each box shall be lined with a sealed glassine or wax paper or other suitable liner material to prevent contents from sifting, and for protection against atmospheric or contaminating conditions. Each box shall be closed in accordance with the appendix of the box specification.

5.1.1.2 *Fifty pounds.* Unless otherwise specified, the contents of bulk dishwashing compound, packed in fiberboard containers shall be completely enclosed in sealed, but recloseable, polyethylene bags within the fiberboard container. The film shall conform to L-P-378, type I, grade A, finish 1, with all seams accomplished by a suitable heat-seal method, which will result in compliance with the heat-seal strength specified in L-P-378. The bag shall be closed by knotting or other suitable means. The thickness of the film for bags shall be not less than 0.002 inch (2.0 mils plus or minus 20 percent). Dishwashing compound, packaged as above, shall be packed in rectangular-shaped boxes conforming to type I or II, class 1, styles FOL, SFF, or CSSC of PPP-B-636. For nonshipboard use, the container may be a paper sack conforming to construction 13 X of UU-S-48.

5.1.1.3 *One-hundred pounds.* The container shall be a fiber drum conforming to type II, grade A of PPP-D-723 with a 3 MIL minimum thick polyethylene liner long enough to wire tie.

5.1.1.4 *Three-hundred pounds.* Container shall be as specified in 5.1.1.3.

5.1.2 *Level B.* The dishwashing compound shall be packaged in accordance with 5.1.1, except that 100 pounds shall be packed in a fiber drum conforming to type III, [illegible] of PPP-D-723. When specified (see 6.2), a liner shall be provided in accordance with 5.1.1.2.

5.1.3 *Level C.* The dishwashing compound shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Paperboard boxes.* Forty-eight pounds of dishwashing compound (24 paperboard boxes), packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RCS, V3c or V3s of PPP-B-636 reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.1.2 *Polyethylene bags and fiber drums.* Dishwashing compound, packaged in polyethylene bags or fiber drums as specified in 5.1.1, shall not require any overpacking.

5.2.2 *Level B.*

5.2.2.1 *Paperboard boxes.* Forty-eight pounds of dishwashing compound (24 pa-

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perboard boxes), packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RCS, type CF or SF, class domestic of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.2.2 *Fifty pounds.* The container shall be a paper shipping sack conforming to construction 9-9X(MB-2) of UU-S-48.

5.2.2.3 *One-hundred pounds.* The shipping container shall be as specified in 5.1.1.3, except type I fiber drums shall be used.

5.2.2.4 *Three-hundred pounds.* The shipping container shall be as specified in 5.1.1.3, except type I fiber drums shall be used.

5.2.3 *Level C.* Dishwashing compound, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages, paper shipping sacks, fiber drums, and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages, paper shipping sacks, fiber drums, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 *Type I, for hard water.* This compound is intended for use in mechanical dishwashing machines supplied with water above 120 p.p.m. hardness, calculated as calcium carbonate (above 7 grains hardness per gallon).

6.1.2 *Type II, for soft water.* This compound is intended for use in mechanical dishwashing machines supplied with water of 0 to 120 p.p.m. hardness, calculated as calcium carbonate (0 to 7 grains hardness per gallon).

6.1.3 This compound is not intended for use in manual cleaning.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type required. (See 1.2.)

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(*d*) Quantity of dishwashing compound per package (see 5.1.1 and 5.1.3).

(*e*) When level B is required, whether liner is required (see 5.1.2).

(*f*) The right is reserved to reject bids from manufacturers who have failed to perform satisfactorily or complete on time a previous contract for dishwashing compound.

6.3 Dishwashing compound is purchased by net weight. It is subject to a possible gain or loss in weight, depending upon packaging, atmospheric or storage conditions, or both. Therefore the net weight in reference to acceptance or delivery should be computed at the time the Government inspector selects samples for lot inspection tests. This time should be specified in the contract or order.

6.4 Level B packaging required for civil agencies.

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MILITARY CUSTODIANS:

Army—GL

Navy—SH

Air Force—69

Review activities:

Army—GL, MD

Navy—SH

Air Force—69

User activity:

Navy—MC, YD

Preparing activity:

Navy—SH

Orders for this publication are to be placed with cting as an agent for the See section 2 of this specification to obtain extra copies and other documents referenced herein.

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